Supplement B - Simulation Atlas

Extended sensitivity analyses for The Link Function Problem in Psychological Interaction Testing

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1 Purpose and scope

Supplement B extends the three manuscript simulations by varying selected parameters while keeping the true interaction equal to zero on the generating scale. It uses one-dimensional sensitivity slices and small main-effect surfaces rather than a full Cartesian design.

Reference scenarios retain the generating settings of the corresponding manuscript simulations. In every scenario, a rejection under the matched generating model estimates the nominal generating-scale rejection rate; under another scale, it is a **pseudo-interaction detection rate**. For forced choice, these primary rates use finite tests from fits passing the minimal lme4 fit-quality checks. Apparent rejection rates among flagged fits are reported separately. Publication runs request 3,000 Monte Carlo replications.

1.1 How to read the figures

All rejection-rate figures use the same interpretation. The horizontal dashed line at .05 marks nominal alpha. Rates close to .05 under the generating scale provide a calibration check. Elevated rates under an alternative fitted scale show how often the scale difference becomes a statistically significant product term at the stated sample size and design. Error bars are shown for the reference scenarios. The sensitivity plots omit error bars to keep the larger set of scenarios readable; Wilson 95% intervals are available in the numerical tables and in `data/atlas-summary.csv`.

The deterministic pseudo-interaction reported in the tables is calculated from expected values before sampling. For forced choice and the other mixed-model family, it is evaluated at random intercept 0, not as a marginal population-average probability. It describes the interaction contrast implied by evaluating the generating pattern on the fitted scale. The Monte Carlo rejection rate adds sampling variability, standard errors, and sample size to that deterministic discrepancy.

2 Atlas design

Table 1: Structure of the simulation atlas. Each family contains reference scenarios plus prespecified sensitivity slices.

Manuscript case	Generating scenarios	Sensitivity slices
Simulation 1: Forced-choice accuracy	24	Main effects; sample size; intercept; trials; chance level
Simulation 2: Logit vs probit	46	Main effects; sample size; trials; intercept; ICC
Simulation 3: Sum scores	23	Main effects; sample size; scale location; item count; latent dispersion

The one-dimensional slices change one named parameter while holding the remaining settings at the central reference scenario. The main-effect surfaces vary the two additive main-effect coefficients jointly. The actual coefficients are reported on the figure axes and in the table below.

Table 2: Generating-scale coefficients used in the main-effect surfaces. Coefficients are reported on the scale that generated each scenario.

Manuscript case	Generating link/scale	First main effect	First coefficient values	Second main effect	Second coefficient values
Simulation 1: Forced-choice accuracy	Chance-corrected logit	Age coefficient	0.30, 0.60, 0.90	Group coefficient	-1.35, -0.90, -0.45
Simulation 2: Logit vs probit	Logit	Group coefficient	-2.475, -1.650, -0.825	Condition coefficient	-2.475, -1.650, -0.825
Simulation 2: Logit vs probit	Probit	Group coefficient	-1.50, -1.00, -0.50	Condition coefficient	-1.50, -1.00, -0.50
Simulation 3: Sum scores	Latent generating scale	Predictor coefficient	0.425, 0.850, 1.275	Group coefficient	-1.35, -0.90, -0.45

For Simulation 2, logit and probit coefficients are on different generating-link scales. As in the manuscript simulation, the logit coefficients are scaled by 1.65 relative to the probit reference coefficients so that the two generating scenarios have similar cell probabilities. The figures therefore report the actual coefficient values for each generating link instead of a common relative index.

3 Simulation 1: Forced-choice accuracy and the non-zero chance floor

Forced-choice accuracy is generated as trial-level Bernoulli responses from a chance-corrected binomial logit mixed model with a .50 floor, a subject random intercept, and no age-by-group product term on the conditional chance-corrected-logit scale. The subject random-intercept SD corresponds to latent-scale ICC .30 under the usual logistic convention. The fitted models are a matched chance-corrected-logit GLMM using `psyphy::mafc.logit`, standard binomial-logit and binomial-probit GLMMs, and a Gaussian identity LMM. All four use the same `(1 | id)` structure.

3.1 Reference scenarios

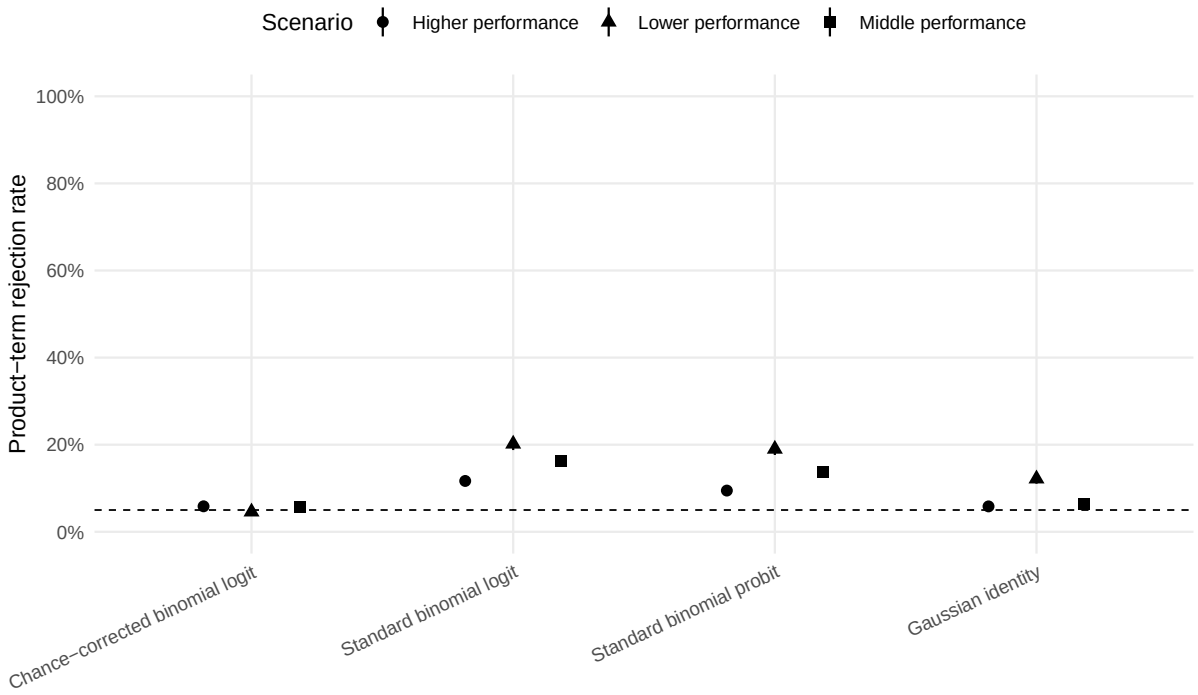


Figure 1: Forced-choice reference scenarios. The product term is zero on the conditional chance-corrected-logit generating scale. Rates use fits passing the minimal fit-quality checks. The dashed line marks alpha = .05.

Table 3: Forced-choice reference scenarios. The deterministic pseudo-interaction is evaluated on each fitted scale before sampling.

Scenario	Generating link/scale	Fitted model/scale	Rate type	Rejection rate	95% CI	Deterministic pseudo-interaction	Flagged fits	Fit-problem rate	Apparent rejection rate among flagged fits
Lower performance	chance-corrected logit	Chance-corrected binomial logit	Nominal generating-scale rejection rate	4.6%	[3.7%, 5.6%]	0.000	1209	40.3%	90.1%
Lower performance	chance-corrected logit	Gaussian identity	Pseudo-interaction detection rate	12.2%	[11.0%, 13.4%]	-0.077	0	0.0%	–
Lower performance	chance-corrected logit	Standard binomial logit	Pseudo-interaction detection rate	20.2%	[18.8%, 21.6%]	-0.453	0	0.0%	–
Lower performance	chance-corrected logit	Standard binomial probit	Pseudo-interaction detection rate	19.0%	[17.6%, 20.4%]	-0.262	0	0.0%	–
Middle performance	chance-corrected logit	Chance-corrected binomial logit	Nominal generating-scale rejection rate	5.8%	[4.8%, 7.0%]	0.000	1270	42.3%	91.0%
Middle performance	chance-corrected logit	Gaussian identity	Pseudo-interaction detection rate	6.4%	[5.6%, 7.3%]	-0.036	0	0.0%	–
Middle performance	chance-corrected logit	Standard binomial logit	Pseudo-interaction detection rate	16.2%	[14.9%, 17.5%]	-0.473	0	0.0%	–

Table 3: Forced-choice reference scenarios. The deterministic pseudo-interaction is evaluated on each fitted scale before sampling.

Scenario	Generating link/scale	Fitted model/scale	Rate type	Rejection rate	95% CI	Deterministic pseudo-interaction	Flagged fits	Fit-problem rate	Apparent rejection rate among flagged fits
Middle performance	chance-corrected logit	Standard binomial probit	Pseudo-interaction detection rate	13.7%	[12.5%, 15.0%]	-0.243	69	2.3%	95.7%
Higher performance	chance-corrected logit	Chance-corrected binomial logit	Nominal generating-scale rejection rate	5.9%	[4.8%, 7.1%]	0.000	1308	43.6%	95.5%
Higher performance	chance-corrected logit	Gaussian identity	Pseudo-interaction detection rate	5.8%	[5.0%, 6.7%]	0.028	0	0.0%	–
Higher performance	chance-corrected logit	Standard binomial logit	Pseudo-interaction detection rate	11.7%	[10.6%, 12.9%]	-0.396	1	0.0%	100.0%
Higher performance	chance-corrected logit	Standard binomial probit	Pseudo-interaction detection rate	9.5%	[8.5%, 10.6%]	-0.153	105	3.5%	81.9%

The forced-choice atlas must be regenerated before numerical ranges are reported for the revised mixed-model design. The saved atlas results from the earlier fixed-effect design are not valid for this specification.

3.2 One-dimensional sensitivity slices

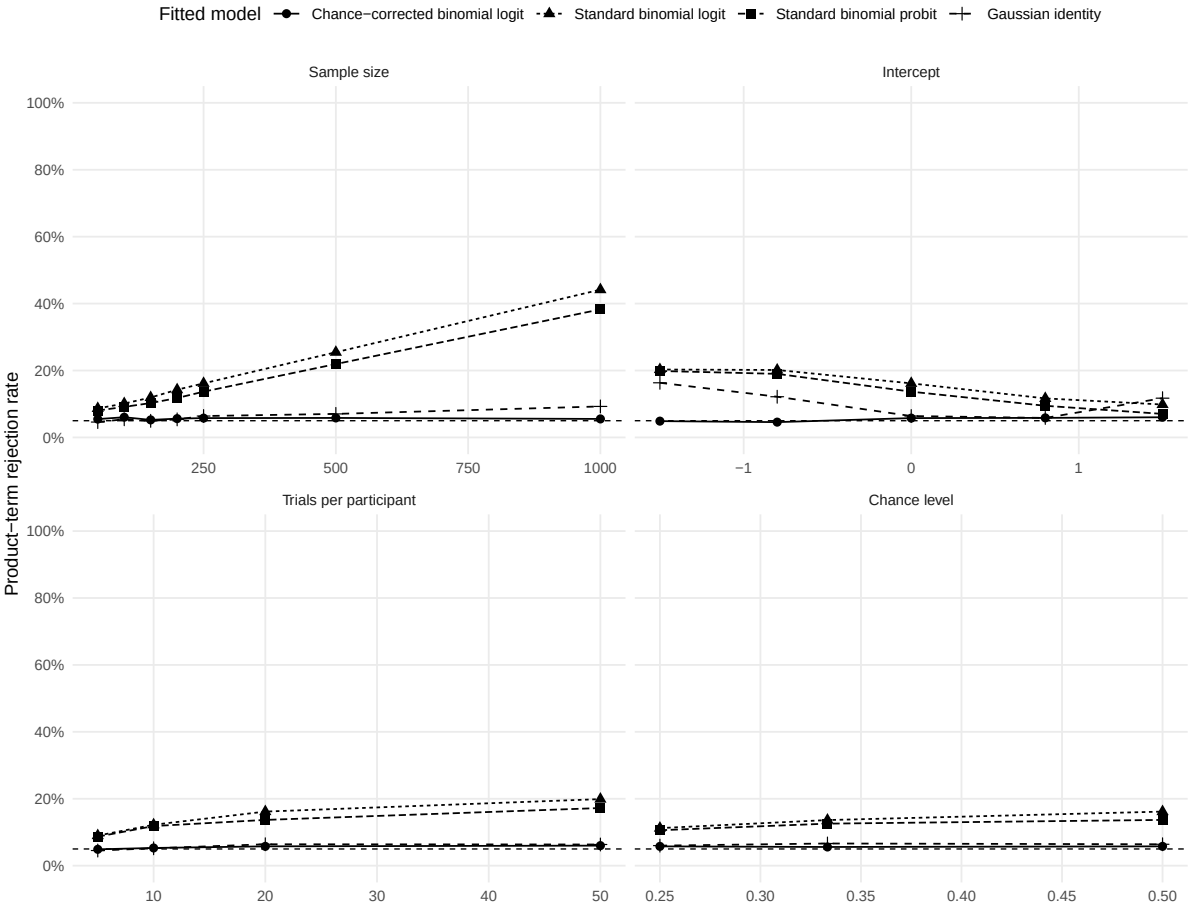


Figure 2: Forced-choice sensitivity slices. One parameter varies at a time around the central reference scenario. The dashed line marks alpha = .05.

Table 4: Range of rejection rates across each forced-choice sensitivity slice.

Slice	Generating link/scale	Fitted model/scale	Minimum rejection rate	Maximum rejection rate
Chance level	chance-corrected logit	Chance-corrected binomial logit	5.6%	5.8%
Intercept	chance-corrected logit	Chance-corrected binomial logit	4.6%	6.0%
Sample size	chance-corrected logit	Chance-corrected binomial logit	5.2%	6.0%
Trials per participant	chance-corrected logit	Chance-corrected binomial logit	4.9%	6.0%
Chance level	chance-corrected logit	Standard binomial logit	11.2%	16.2%
Intercept	chance-corrected logit	Standard binomial logit	9.8%	20.3%
Sample size	chance-corrected logit	Standard binomial logit	8.7%	44.2%
Trials per participant	chance-corrected logit	Standard binomial logit	9.0%	19.9%
Chance level	chance-corrected logit	Standard binomial probit	10.6%	13.7%
Intercept	chance-corrected logit	Standard binomial probit	7.0%	19.9%
Sample size	chance-corrected logit	Standard binomial probit	8.0%	38.3%
Trials per participant	chance-corrected logit	Standard binomial probit	8.7%	17.2%
Chance level	chance-corrected logit	Gaussian identity	6.0%	6.6%
Intercept	chance-corrected logit	Gaussian identity	5.8%	16.3%
Sample size	chance-corrected logit	Gaussian identity	4.6%	9.3%
Trials per participant	chance-corrected logit	Gaussian identity	4.5%	6.4%

The slices separate distinct reasons why a deterministic scale discrepancy may or may not

become a statistical interaction. The intercept and the chance level change the curvature relevant to the fitted link. Sample size and number of trials change how precisely that discrepancy is estimated. The figure therefore distinguishes the existence of a pseudo-interaction on the fitted scale from the probability that a product-term test detects it.

3.3 Main-effect surface

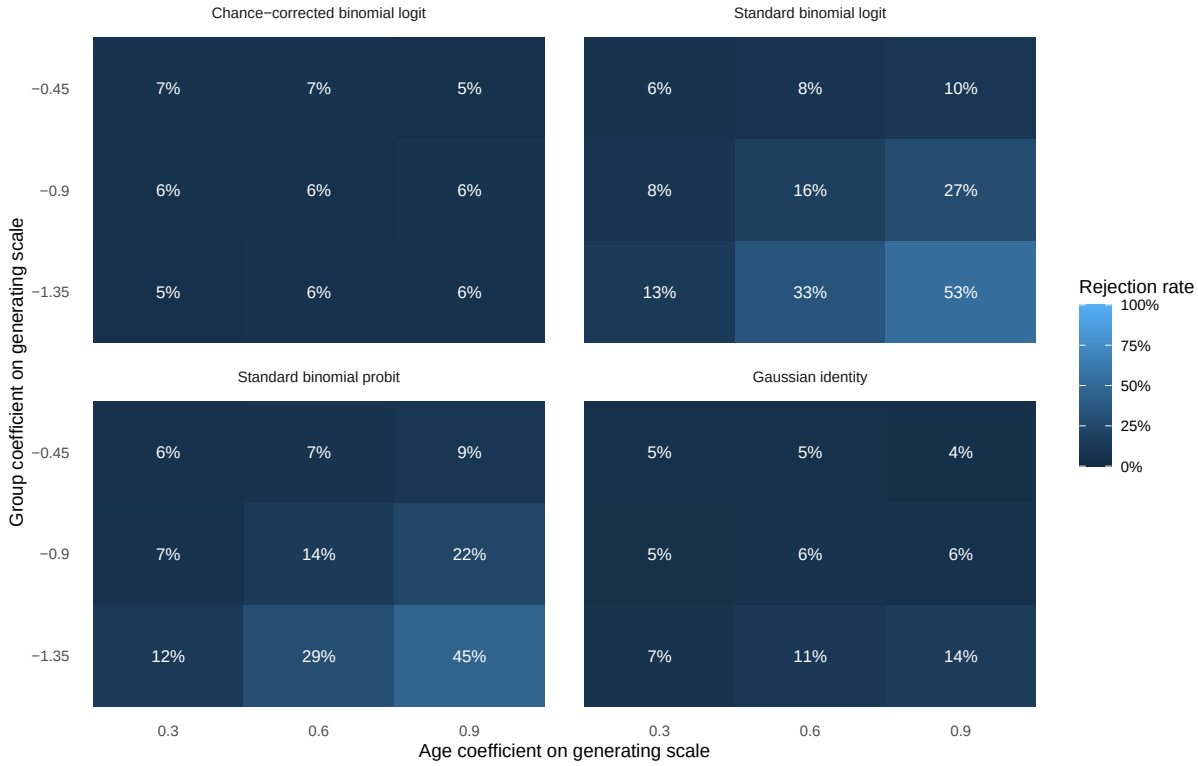


Figure 3: Forced-choice main-effect surface. Axes report the age and group coefficients on the chance-corrected logit generating scale. Cell labels are product-term rejection rates.

The surface directly targets the mechanism emphasized in the manuscript. Two additive main effects can jointly move observations through regions where equal differences on the generating scale map to unequal differences on the fitted scale. Larger main effects can therefore increase the deterministic pseudo-interaction even though the generating age-by-group term remains zero.

4 Simulation 2: Within-family link choice, logit versus probit

Repeated-trial random-intercept GLMMs cross logit and probit generating and fitted links. Both use the binomial family and the same 0-to-1 bounds, but impose additivity on different scales; the figures and Table 1 show the sensitivity design.

4.1 Reference scenarios

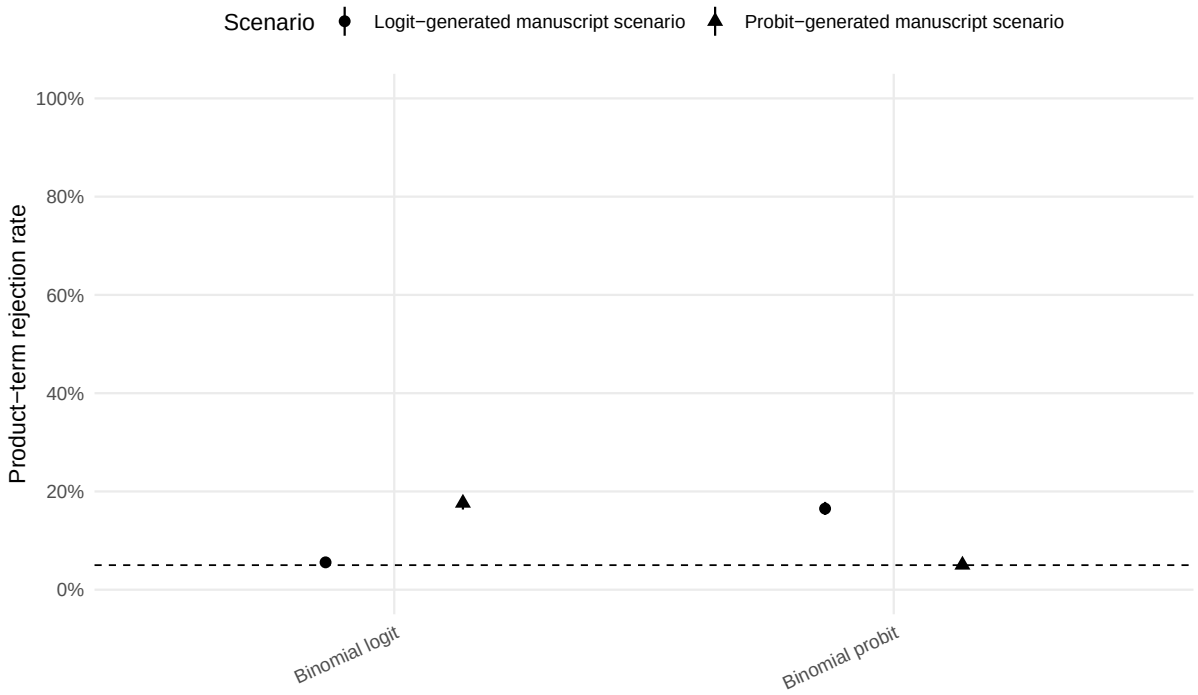


Figure 4: Logit-probit reference scenarios. The product term is zero on the generating link scale. The dashed line marks alpha = .05.

Table 5: Within-family reference scenarios for logit and probit generating links.

Scenario	Generating link/scale	Fitted model/scale	Rate type	Rejection rate	95% CI	Deterministic pseudo-interaction
Logit-generated manuscript scenario	logit	Binomial logit	Nominal generating-scale rejection rate	5.6%	[4.8%, 6.4%]	0.000

Table 5: Within-family reference scenarios for logit and probit generating links.

Scenario	Generating link/scale	Fitted model/scale	Rate type	Rejection rate	95% CI	Deterministic pseudo-interaction
Logit-generated manuscript scenario	logit	Binomial probit	Pseudo-interaction detection rate	16.5%	[15.2%, 17.9%]	-0.112
Probit-generated manuscript scenario	probit	Binomial logit	Pseudo-interaction detection rate	17.6%	[16.3%, 19.0%]	0.216
Probit-generated manuscript scenario	probit	Binomial probit	Nominal generating-scale rejection rate	5.0%	[4.3%, 5.9%]	0.000

Across the atlas, matched logit/probit fits remain near the nominal rejection rate. The wrong-link rows range from 4.4% to 82.3%. The range is wider than the reference scenarios alone suggest because the atlas deliberately varies information and the region of the latent scale. The result remains conditional on these declared slices and should not be read as a general frequency for logit-probit disagreement.

4.2 One-dimensional sensitivity slices

Table 6: Range of rejection rates across each within-family sensitivity slice.

Slice	Generating link/scale	Fitted model/scale	Minimum rejection rate	Maximum rejection rate
Intercept	logit	Binomial logit	5.1%	5.6%
Latent ICC	logit	Binomial logit	4.9%	5.9%
Sample size	logit	Binomial logit	4.4%	5.7%
Trials per cell	logit	Binomial logit	4.4%	5.6%
Intercept	probit	Binomial logit	8.0%	46.2%
Latent ICC	probit	Binomial logit	8.6%	82.3%
Sample size	probit	Binomial logit	6.6%	41.5%

Table 6: Range of rejection rates across each within-family sensitivity slice.

Slice	Generating link/scale	Fitted model/scale	Minimum rejection rate	Maximum rejection rate
Trials per cell	probit	Binomial logit	7.9%	32.2%
Intercept	logit	Binomial probit	8.2%	44.1%
Latent ICC	logit	Binomial probit	9.2%	78.7%
Sample size	logit	Binomial probit	7.4%	38.5%
Trials per cell	logit	Binomial probit	8.2%	31.0%
Intercept	probit	Binomial probit	4.7%	5.7%
Latent ICC	probit	Binomial probit	4.8%	5.4%
Sample size	probit	Binomial probit	4.5%	5.6%
Trials per cell	probit	Binomial probit	4.6%	5.0%

Sample size and trial count mainly change how detectable the small systematic discrepancy between the links becomes. The intercept and the main-effect magnitudes change the discrepancy itself by moving cell probabilities into regions where logit and probit curvature differs more. ICC changes the information structure of the repeated observations while leaving the fixed generating interaction at zero.

4.3 Main-effect surface

The matched-link panels provide the calibration reference. The off-diagonal panels show when the small difference between two standard binomial links becomes consequential for an interaction claim. This is the narrow point of the within-family example: logit and probit can remain very similar for prediction while defining sufficiently different additivity baselines to change a product-term conclusion under high information or strong curvature.

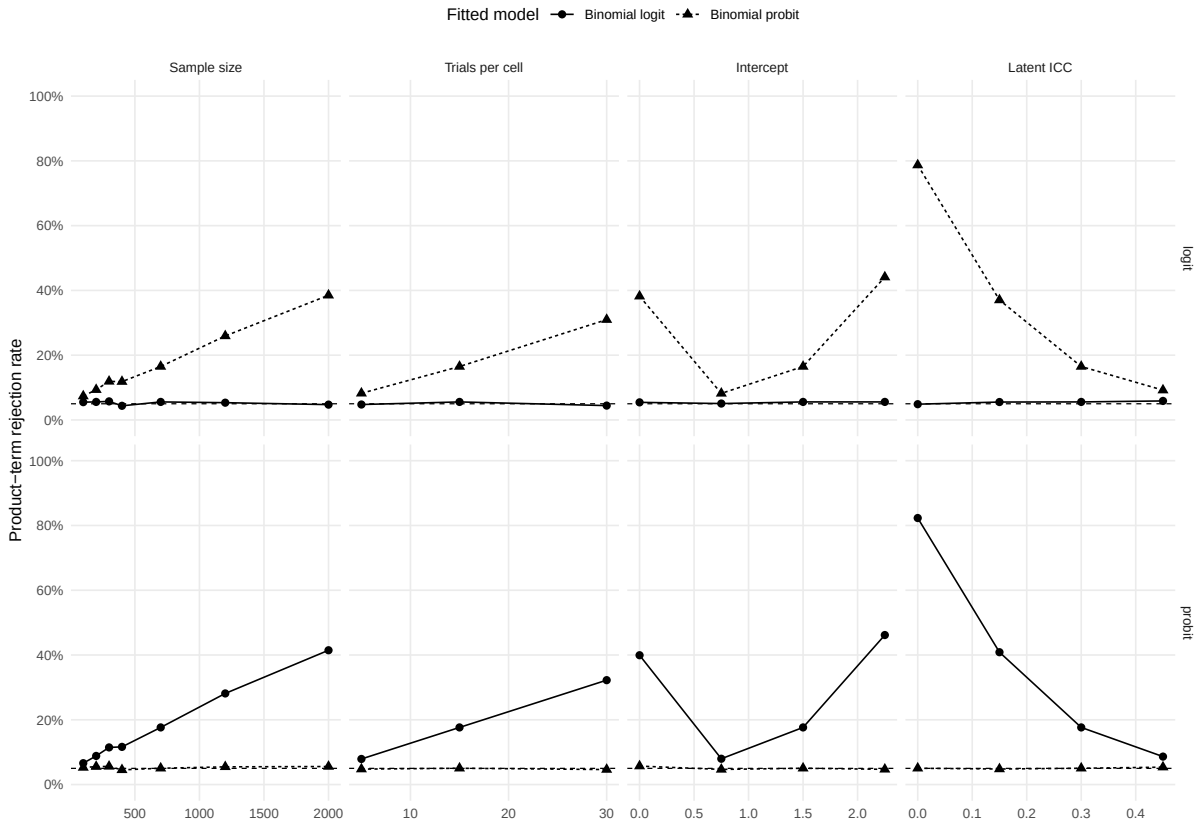


Figure 5: Logit-probit sensitivity slices. Rows identify the generating link. Within each panel the fitted logit and probit models are compared as one parameter varies around the reference scenario.

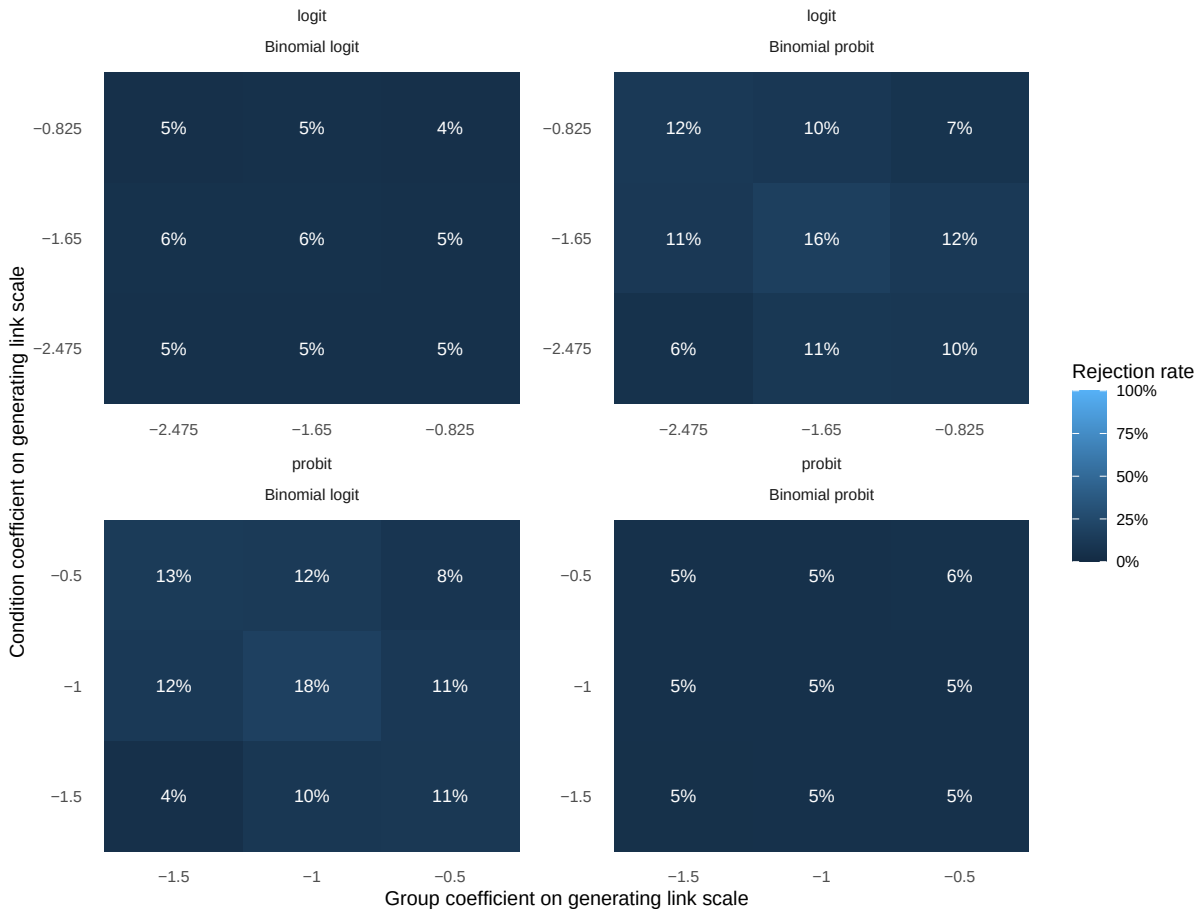


Figure 6: Within-family main-effect surfaces. Axes report the group and condition coefficients on each generating-link scale. Panels identify the generating and fitted links. Cell labels are product-term rejection rates.

5 Simulation 3: Sum scores

Sum-score data are generated on a latent continuous scale with no x-by-group interaction, converted to ordinal item responses, and summed. Analyses compare the latent generating scale, observed sum-score identity model, and Gaussian-probit bounded-score sensitivity model; the figures and Table 1 show the sensitivity design.

5.1 Reference scenarios

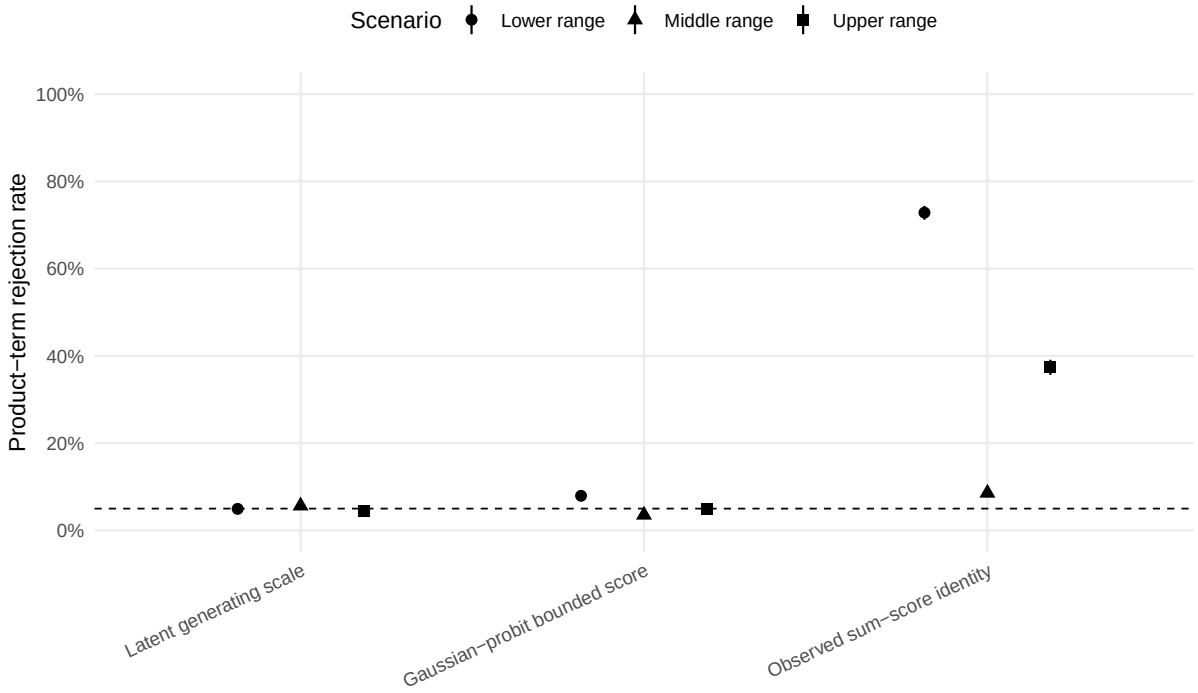


Figure 7: Sum-score reference scenarios. The product term is zero on the latent generating scale. The dashed line marks alpha = .05.

Table 7: Sum-score reference scenarios. The latent generating scale is available only in simulation.

Scenario	Generating link/scale	Fitted model/scale	Rate type	Rejection rate	95% CI	Deterministic pseudo-interaction
Lower range	graded logistic latent scale	Gaussian-probit bounded score	Pseudo-interaction detection rate	7.9%	[7.0%, 9.0%]	-0.056
Lower range	graded logistic latent scale	Latent generating scale	Nominal generating-scale rejection rate	4.9%	[4.2%, 5.8%]	0.000
Lower range	graded logistic latent scale	Observed sum-score identity	Pseudo-interaction detection rate	72.8%	[71.2%, 74.4%]	-2.034
Middle range	graded logistic latent scale	Gaussian-probit bounded score	Pseudo-interaction detection rate	3.5%	[2.9%, 4.3%]	-0.010
Middle range	graded logistic latent scale	Latent generating scale	Nominal generating-scale rejection rate	5.6%	[4.9%, 6.5%]	0.000
Middle range	graded logistic latent scale	Observed sum-score identity	Pseudo-interaction detection rate	8.6%	[7.6%, 9.6%]	-0.696
Upper range	graded logistic latent scale	Gaussian-probit bounded score	Pseudo-interaction detection rate	4.8%	[4.1%, 5.6%]	0.027
Upper range	graded logistic latent scale	Latent generating scale	Nominal generating-scale rejection rate	4.5%	[3.8%, 5.3%]	0.000
Upper range	graded logistic latent scale	Observed sum-score identity	Pseudo-interaction detection rate	37.4%	[35.7%, 39.1%]	1.570

Across the atlas, the latent generating scale rejects from 4.2% to 6.0%. The observed sum-score identity model ranges from 2.4% to 82.1%, while the Gaussian-probit bounded-score sensitivity model ranges from 3.1% to 11.9%. The latter remains a model of the manifest total. It does

not recover the latent construct and should be read as a sensitivity analysis for the scale on which additivity is imposed.

5.2 One-dimensional sensitivity slices

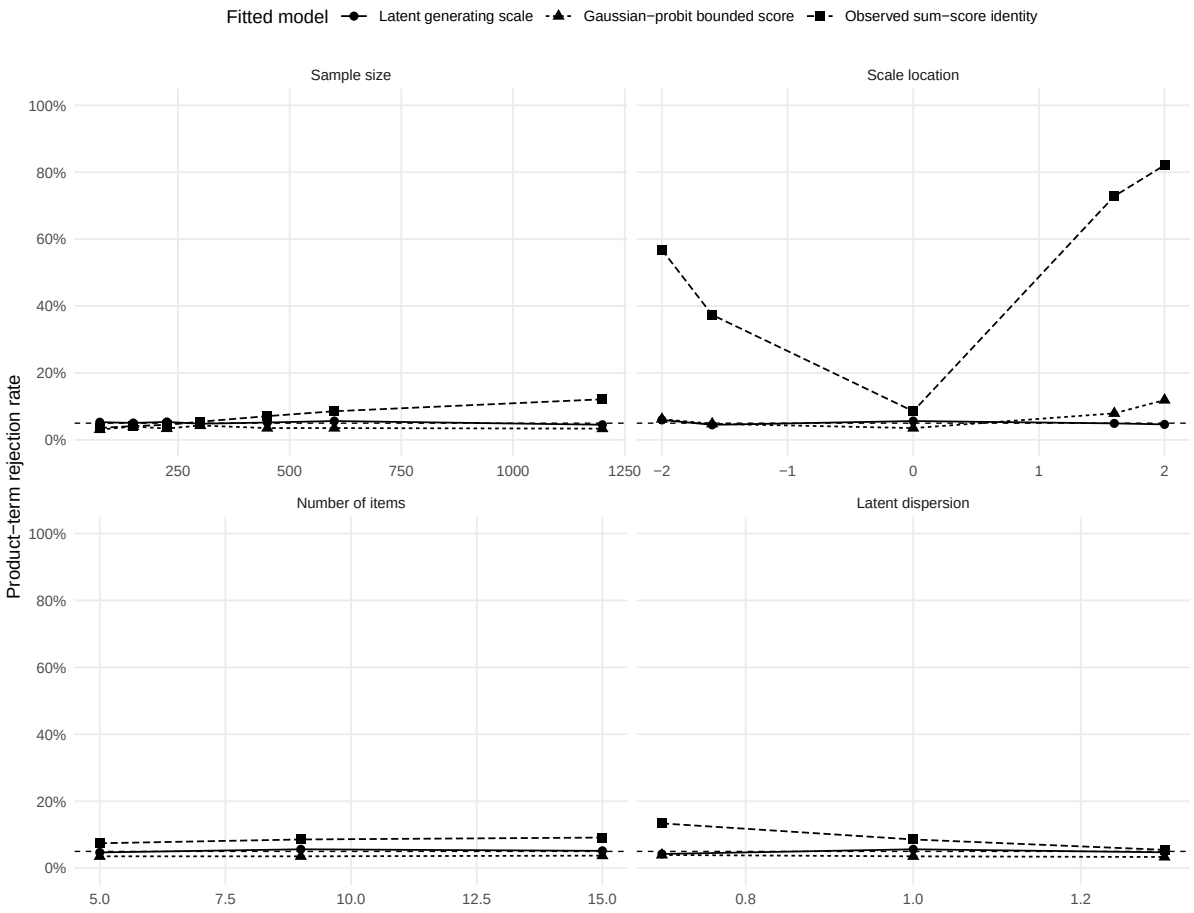


Figure 8: Sum-score sensitivity slices. One parameter varies at a time around the central reference scenario. The dashed line marks alpha = .05.

Table 8: Range of rejection rates across each sum-score sensitivity slice.

Slice	Generating link/scale	Fitted model/scale	Minimum rejection rate	Maximum rejection rate
Latent dispersion	graded logistic latent scale	Latent generating scale	4.2%	5.6%

Table 8: Range of rejection rates across each sum-score sensitivity slice.

Slice	Generating link/scale	Fitted model/scale	Minimum rejection rate	Maximum rejection rate
Number of items	graded logistic latent scale	Latent generating scale	4.7%	5.6%
Sample size	graded logistic latent scale	Latent generating scale	4.5%	5.6%
Scale location	graded logistic latent scale	Latent generating scale	4.5%	6.0%
Latent dispersion	graded logistic latent scale	Gaussian-probit bounded score	3.3%	3.9%
Number of items	graded logistic latent scale	Gaussian-probit bounded score	3.5%	3.7%
Sample size	graded logistic latent scale	Gaussian-probit bounded score	3.1%	4.3%
Scale location	graded logistic latent scale	Gaussian-probit bounded score	3.5%	11.9%
Latent dispersion	graded logistic latent scale	Observed sum-score identity	5.5%	13.4%
Number of items	graded logistic latent scale	Observed sum-score identity	7.4%	9.1%
Sample size	graded logistic latent scale	Observed sum-score identity	3.6%	12.1%
Scale location	graded logistic latent scale	Observed sum-score identity	8.6%	82.1%

The scale-location slice changes how strongly the bounded sum score compresses the latent dimension. Item count and latent dispersion change the mapping between the latent variable and the manifest total. Sample size changes the probability that any resulting fitted-scale interaction is detected. These are distinct components of the sum-score problem and should not be reduced to a generic floor-or-ceiling warning.

5.3 Main-effect surface

The surface shows how the mapping from latent values to a bounded total can turn two additive latent effects into a non-additive pattern on the manifest score. The latent benchmark remains the calibration reference because its product term is zero by construction.

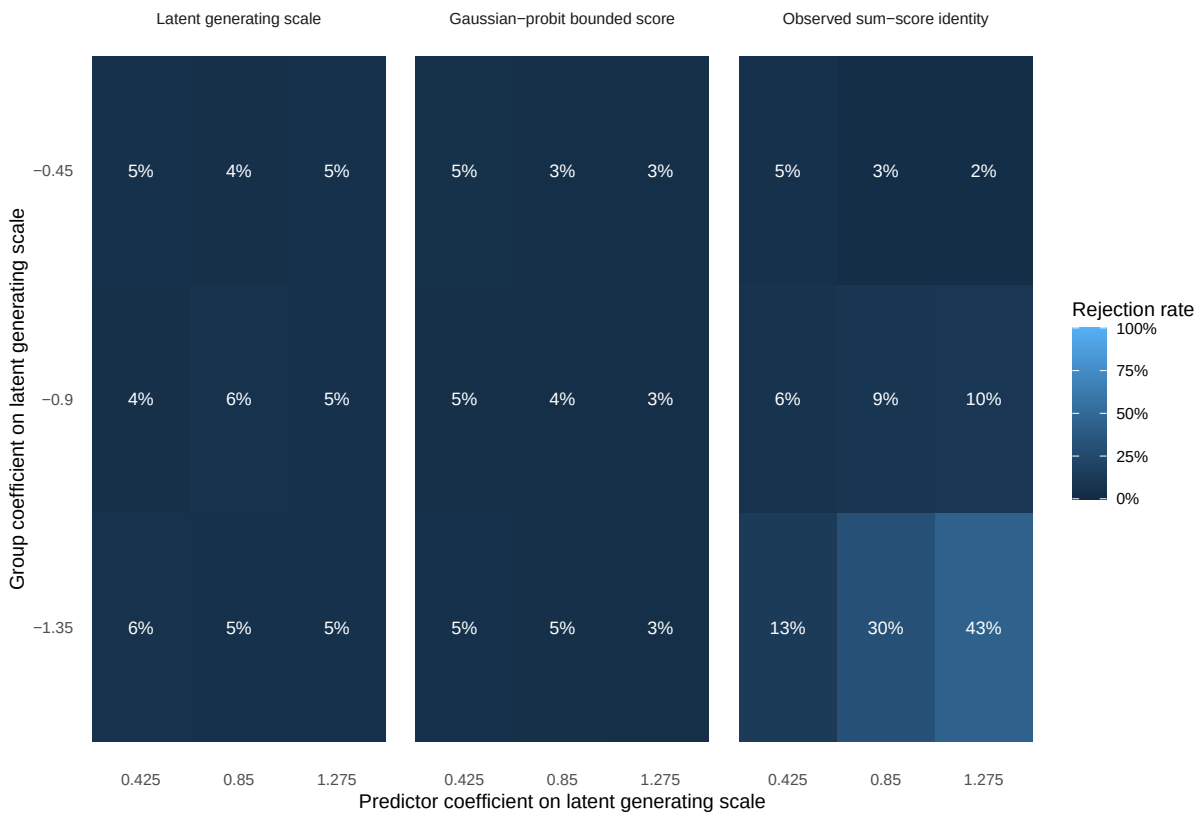


Figure 9: Sum-score main-effect surface. Axes report the predictor and group coefficients on the latent generating scale. Cell labels are product-term rejection rates.

6 Diagnostic atlas

The diagnostic analyses cover the forced-choice chance-floor case and the probit-generated, logit-fitted repeated-trial case. The forced-choice diagnostic uses the same trial-level random-intercept DGP and paired standard-logit and chance-corrected-logit GLMMs as the revised manuscript scenario. They examine how pseudo-interaction risk and diagnostic detection change with sample size and intercept. The forced-choice diagnostic atlas must be regenerated before its numerical values are interpreted.

Diagnostics are applied to the **misspecified interaction model**, matching the manuscript's post-estimation question: once the fitted model has produced the product-term finding, does that same model show enough evidence of misspecification to warn the analyst? AIC compares prespecified candidate mean models with the same structural interaction formula. Positive delta AIC, defined as AIC of the misspecified model minus AIC of the generating model, favors the generating link.

Table 9: Diagnostics used for each diagnostic case.

Diagnostic case	Diagnostics used
Forced-choice chance-floor case	AIC favors generating link; DHARMa uniformity; DHARMa dispersion; DHARMa residual quantiles over fitted values; DHARMa residual quantiles over focal predictor; Pregibon-style added-term link check
Probit-generated, logit-fitted case	AIC favors generating link; DHARMa uniformity; DHARMa dispersion; DHARMa residual distribution across design cells; Pregibon-style added-term link check

Checks not listed were not applicable to the design. In particular, the design-cell residual check is used only for the categorical repeated-trial case, whereas residual quantile checks over a focal predictor require a continuous predictor.

6.1 Detection across sensitivity slices

The rates in this figure answer different questions. Pseudo-interaction detection is the probability that the misspecified fitted scale yields a significant product term. Pregibon-style detection is the probability that the added-term link check flags a departure. AIC reports how often the generating link has the lower AIC within the specified candidate pair. The curves therefore need not be complements of one another. Diagnostic misalignment occurs when the interaction conclusion changes more readily than the diagnostic workflow signals a problem.

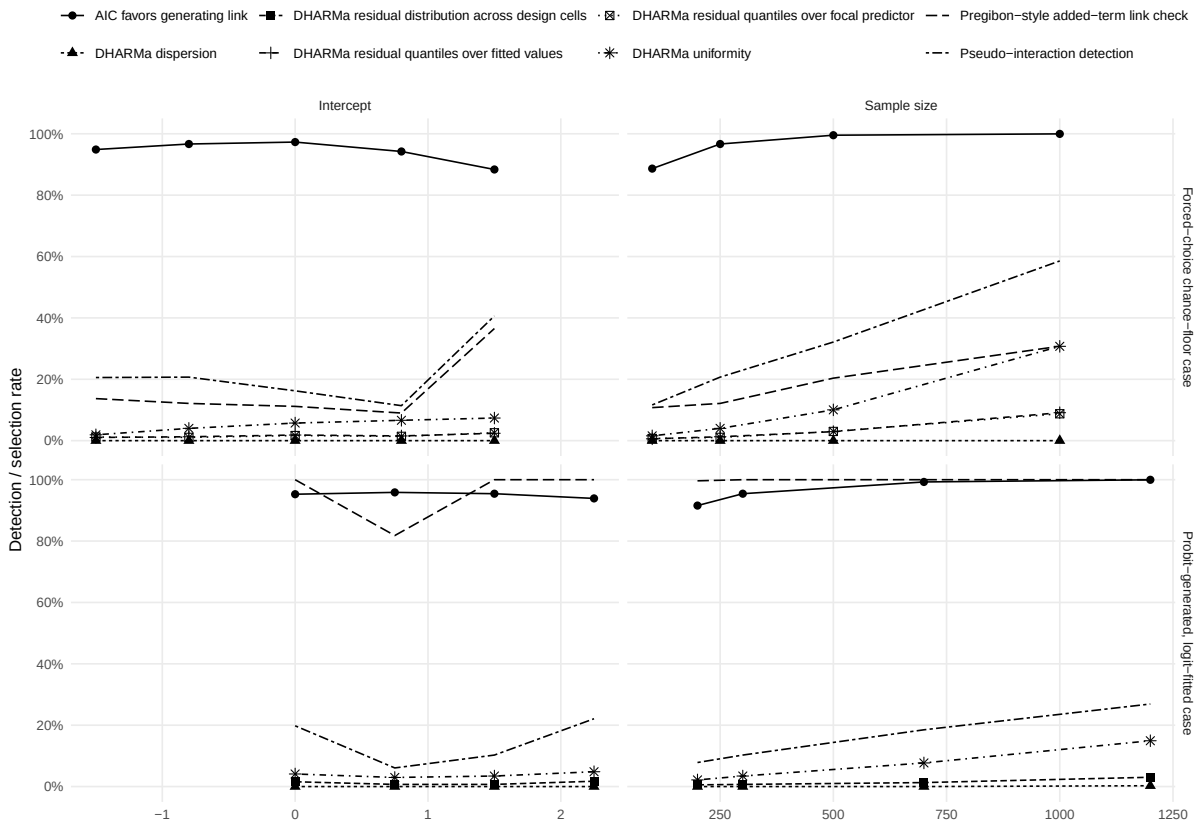


Figure 10: Diagnostic atlas. Pseudo-interaction detection is shown together with each diagnostic or candidate-model comparison. Sample-size and scale-location slices are shown separately.

6.2 Strength of the AIC comparison

Table 10: AIC comparison across the diagnostic atlas. Positive delta AIC favors the generating link. The within-two column reports the proportion of finite comparisons with absolute delta AIC below 2.

Case	Setting	Pseudo-interaction detection	AIC favors generating link	Median delta AIC	Delta AIC IQR	Absolute delta AIC < 2
Forced-choice chance-floor case	Sample size = 100	11.6%	88.7%	5.24	[2.14, 8.62]	18.3%
Forced-choice chance-floor case	Sample size = 250	20.7%	96.7%	13.17	[8.36, 18.41]	4.0%
Forced-choice chance-floor case	Sample size = 500	32.1%	99.5%	27.70	[20.61, 34.85]	0.4%
Forced-choice chance-floor case	Sample size = 1000	58.6%	100.0%	55.51	[44.85, 65.60]	0.0%
Probit-generated, logit-fitted case	Sample size = 200	7.8%	91.6%	3.92	[1.99, 5.77]	23.1%
Probit-generated, logit-fitted case	Sample size = 300	10.3%	95.4%	6.04	[3.72, 8.46]	11.1%
Probit-generated, logit-fitted case	Sample size = 700	18.5%	99.3%	13.76	[10.02, 17.48]	1.3%
Probit-generated, logit-fitted case	Sample size = 1200	26.9%	100.0%	23.54	[18.83, 28.28]	0.1%
Forced-choice chance-floor case	Intercept = -0.80	20.7%	96.7%	13.17	[8.36, 18.41]	4.0%
Forced-choice chance-floor case	Intercept = -1.50	20.6%	94.9%	10.42	[5.75, 15.57]	7.2%
Forced-choice chance-floor case	Intercept = 0.00	16.2%	97.3%	13.18	[8.77, 18.02]	3.3%

Table 10: AIC comparison across the diagnostic atlas. Positive delta AIC favors the generating link. The within-two column reports the proportion of finite comparisons with absolute delta AIC below 2.

Case	Setting	Pseudo-interaction detection	AIC favors generating link	Median delta AIC	Delta AIC IQR	Absolute delta AIC < 2
Forced-choice chance-floor case	Intercept = 0.80	11.4%	94.2%	9.28	[5.46, 13.10]	8.2%
Forced-choice chance-floor case	Intercept = 1.50	40.6%	88.4%	5.31	[2.45, 8.16]	16.4%
Probit-generated, logit-fitted case	Intercept = 1.50	10.3%	95.4%	6.04	[3.72, 8.46]	11.1%
Probit-generated, logit-fitted case	Intercept = 0.00	19.8%	95.3%	6.68	[4.03, 9.34]	10.3%
Probit-generated, logit-fitted case	Intercept = 0.75	6.1%	95.9%	5.56	[3.34, 7.70]	12.9%
Probit-generated, logit-fitted case	Intercept = 2.25	22.1%	93.9%	6.50	[3.58, 9.41]	12.3%

The AIC win rate is reported together with the magnitude of the difference because these describe different situations. A generating link that wins frequently by a very small margin is weaker evidence than one that wins by a large margin. As in the manuscript, these comparisons rank a limited candidate set. They do not identify the substantively correct scale by themselves.

7 Scope and interpretation

The sensitivity results show whether the reference scenarios reflect broader patterns or depend on a narrow configuration. They also distinguish parameters that change the **deterministic scale discrepancy**, such as the intercept and main-effect magnitude, from parameters that mainly change **detectability**, such as sample size and trial count. The diagnostic analyses place pseudo-interaction detection and model-checking performance on the same sensitivity slices.

The atlas does not estimate how often link misspecification occurs in psychological research. It does not cover every plausible outcome family, link, random-effects structure, measurement model, or predictor distribution. It also does not turn model diagnostics into a decision rule for the scientifically correct scale. Its role is narrower: to show how the mechanisms developed in the paper change across a declared set of nearby scenarios.

8 Numerical appendix

The tables below provide scenario-level results for the core sensitivity analyses. Publication runs request 3,000 Monte Carlo replications.

8.1 Forced-choice accuracy

Table 11: Scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit-problem rate	Apparent rejection rate among flagged fits
FC-001	paper anchor	Lower performance	chance-corrected logit	Chance-corrected binomial logit	4.6%	0.000	40.3%	90.1%
FC-001	paper anchor	Lower performance	chance-corrected logit	Gaussian identity	12.2%	-0.077	0.0%	–
FC-001	paper anchor	Lower performance	chance-corrected logit	Standard binomial logit	20.2%	-0.453	0.0%	–
FC-001	paper anchor	Lower performance	chance-corrected logit	Standard binomial probit	19.0%	-0.262	0.0%	–
FC-002	paper anchor	Middle performance	chance-corrected logit	Chance-corrected binomial logit	5.8%	0.000	42.3%	91.0%
FC-002	paper anchor	Middle performance	chance-corrected logit	Gaussian identity	6.4%	-0.036	0.0%	–
FC-002	paper anchor	Middle performance	chance-corrected logit	Standard binomial logit	16.2%	-0.473	0.0%	–
FC-002	paper anchor	Middle performance	chance-corrected logit	Standard binomial probit	13.7%	-0.243	2.3%	95.7%

Table 11: Scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit-problem rate	Apparent rejection rate among flagged fits
FC-003	paper anchor	Higher performance	chance-corrected logit	Chance-corrected binomial logit	5.9%	0.000	43.6%	95.5%
FC-003	paper anchor	Higher performance	chance-corrected logit	Gaussian identity	5.8%	0.028	0.0%	–
FC-003	paper anchor	Higher performance	chance-corrected logit	Standard binomial logit	11.7%	-0.396	0.0%	100.0%
FC-003	paper anchor	Higher performance	chance-corrected logit	Standard binomial probit	9.5%	-0.153	3.5%	81.9%
FC-004	main effect surface	Age effect x 0.50; group effect x 0.50	chance-corrected logit	Chance-corrected binomial logit	6.7%	0.000	51.4%	93.3%
FC-004	main effect surface	Age effect x 0.50; group effect x 0.50	chance-corrected logit	Gaussian identity	5.4%	-0.007	0.0%	–
FC-004	main effect surface	Age effect x 0.50; group effect x 0.50	chance-corrected logit	Standard binomial logit	6.4%	-0.124	0.0%	–
FC-004	main effect surface	Age effect x 0.50; group effect x 0.50	chance-corrected logit	Standard binomial probit	6.2%	-0.063	0.7%	81.8%
FC-005	main effect surface	Age effect x 1.00; group effect x 0.50	chance-corrected logit	Chance-corrected binomial logit	6.6%	0.000	43.3%	92.9%

Table 11: Scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit-problem rate	Apparent rejection rate among flagged fits
FC-005	main effect surface	Age effect x 1.00; group effect x 0.50	chance-corrected logit	Gaussian identity	5.2%	-0.009	0.0%	–
FC-005	main effect surface	Age effect x 1.00; group effect x 0.50	chance-corrected logit	Standard binomial logit	8.0%	-0.232	0.0%	–
FC-005	main effect surface	Age effect x 1.00; group effect x 0.50	chance-corrected logit	Standard binomial probit	6.8%	-0.113	13.1%	94.9%
FC-006	main effect surface	Age effect x 1.50; group effect x 0.50	chance-corrected logit	Chance-corrected binomial logit	5.3%	0.000	38.5%	92.0%
FC-006	main effect surface	Age effect x 1.50; group effect x 0.50	chance-corrected logit	Gaussian identity	3.9%	-0.009	0.0%	–
FC-006	main effect surface	Age effect x 1.50; group effect x 0.50	chance-corrected logit	Standard binomial logit	9.6%	-0.313	0.0%	–
FC-006	main effect surface	Age effect x 1.50; group effect x 0.50	chance-corrected logit	Standard binomial probit	8.9%	-0.145	43.4%	92.2%
FC-007	main effect surface	Age effect x 0.50; group effect x 1.00	chance-corrected logit	Chance-corrected binomial logit	5.5%	0.000	49.5%	92.9%

Table 11: Scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit-problem rate	Apparent rejection rate among flagged fits
FC-007	main effect surface	Age effect x 0.50; group effect x 1.00	chance-corrected logit	Gaussian identity	4.7%	-0.024	0.0%	–
FC-007	main effect surface	Age effect x 0.50; group effect x 1.00	chance-corrected logit	Standard binomial logit	8.0%	-0.255	0.0%	–
FC-007	main effect surface	Age effect x 0.50; group effect x 1.00	chance-corrected logit	Standard binomial probit	7.5%	-0.136	0.1%	100.0%
FC-008	main effect surface	Age effect x 1.50; group effect x 1.00	chance-corrected logit	Chance-corrected binomial logit	6.4%	0.000	36.0%	89.5%
FC-008	main effect surface	Age effect x 1.50; group effect x 1.00	chance-corrected logit	Gaussian identity	6.1%	-0.034	0.0%	–
FC-008	main effect surface	Age effect x 1.50; group effect x 1.00	chance-corrected logit	Standard binomial logit	26.8%	-0.635	0.0%	–
FC-008	main effect surface	Age effect x 1.50; group effect x 1.00	chance-corrected logit	Standard binomial probit	22.3%	-0.309	25.3%	94.5%
FC-009	main effect surface	Age effect x 0.50; group effect x 1.50	chance-corrected logit	Chance-corrected binomial logit	5.1%	0.000	48.1%	93.6%

Table 11: Scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit-problem rate	Apparent rejection rate among flagged fits
FC-009	main effect surface	Age effect x 0.50; group effect x 1.50	chance-corrected logit	Gaussian identity	7.2%	-0.048	0.0%	–
FC-009	main effect surface	Age effect x 0.50; group effect x 1.50	chance-corrected logit	Standard binomial logit	12.7%	-0.380	0.0%	–
FC-009	main effect surface	Age effect x 0.50; group effect x 1.50	chance-corrected logit	Standard binomial probit	11.5%	-0.210	0.1%	100.0%
FC-010	main effect surface	Age effect x 1.00; group effect x 1.50	chance-corrected logit	Chance-corrected binomial logit	5.8%	0.000	38.5%	91.2%
FC-010	main effect surface	Age effect x 1.00; group effect x 1.50	chance-corrected logit	Gaussian identity	11.0%	-0.073	0.0%	–
FC-010	main effect surface	Age effect x 1.00; group effect x 1.50	chance-corrected logit	Standard binomial logit	32.9%	-0.706	0.0%	–
FC-010	main effect surface	Age effect x 1.00; group effect x 1.50	chance-corrected logit	Standard binomial probit	28.6%	-0.376	1.4%	97.7%
FC-011	main effect surface	Age effect x 1.50; group effect x 1.50	chance-corrected logit	Chance-corrected binomial logit	6.3%	0.000	34.6%	90.8%

Table 11: Scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit-problem rate	Apparent rejection rate among flagged fits
FC-011	main effect surface	Age effect x 1.50; group effect x 1.50	chance-corrected logit	Gaussian identity	14.5%	-0.073	0.0%	–
FC-011	main effect surface	Age effect x 1.50; group effect x 1.50	chance-corrected logit	Standard binomial logit	52.8%	-0.949	0.0%	–
FC-011	main effect surface	Age effect x 1.50; group effect x 1.50	chance-corrected logit	Standard binomial probit	45.1%	-0.481	12.9%	98.4%
FC-012	sample size	N = 50	chance-corrected logit	Chance-corrected binomial logit	5.5%	0.000	33.7%	84.9%
FC-012	sample size	N = 50	chance-corrected logit	Gaussian identity	4.6%	-0.036	0.0%	–
FC-012	sample size	N = 50	chance-corrected logit	Standard binomial logit	8.7%	-0.473	0.1%	0.0%
FC-012	sample size	N = 50	chance-corrected logit	Standard binomial probit	8.0%	-0.243	6.4%	91.7%
FC-013	sample size	N = 100	chance-corrected logit	Chance-corrected binomial logit	6.0%	0.000	38.2%	88.3%
FC-013	sample size	N = 100	chance-corrected logit	Gaussian identity	5.4%	-0.036	0.0%	–
FC-013	sample size	N = 100	chance-corrected logit	Standard binomial logit	10.1%	-0.473	0.0%	–

Table 11: Scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit-problem rate	Apparent rejection rate among flagged fits
FC-013	sample size	N = 100	chance-corrected logit	Standard binomial probit	9.1%	-0.243	6.9%	92.8%
FC-014	sample size	N = 150	chance-corrected logit	Chance-corrected binomial logit	5.2%	0.000	40.3%	89.7%
FC-014	sample size	N = 150	chance-corrected logit	Gaussian identity	5.0%	-0.036	0.0%	–
FC-014	sample size	N = 150	chance-corrected logit	Standard binomial logit	11.9%	-0.473	0.0%	–
FC-014	sample size	N = 150	chance-corrected logit	Standard binomial probit	10.3%	-0.243	5.3%	94.3%
FC-015	sample size	N = 200	chance-corrected logit	Chance-corrected binomial logit	5.6%	0.000	43.7%	91.4%
FC-015	sample size	N = 200	chance-corrected logit	Gaussian identity	5.4%	-0.036	0.0%	–
FC-015	sample size	N = 200	chance-corrected logit	Standard binomial logit	14.3%	-0.473	0.0%	–
FC-015	sample size	N = 200	chance-corrected logit	Standard binomial probit	11.8%	-0.243	3.5%	96.2%
FC-016	sample size	N = 500	chance-corrected logit	Chance-corrected binomial logit	5.8%	0.000	0.0%	–

Table 11: Scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit-problem rate	Apparent rejection rate among flagged fits
FC-016	sample size	N = 500	chance-corrected logit	Gaussian identity	7.0%	-0.036	0.0%	–
FC-016	sample size	N = 500	chance-corrected logit	Standard binomial logit	25.5%	-0.473	0.0%	–
FC-016	sample size	N = 500	chance-corrected logit	Standard binomial probit	21.9%	-0.243	0.0%	–
FC-017	sample size	N = 1000	chance-corrected logit	Chance-corrected binomial logit	5.5%	0.000	0.0%	–
FC-017	sample size	N = 1000	chance-corrected logit	Gaussian identity	9.3%	-0.036	0.0%	–
FC-017	sample size	N = 1000	chance-corrected logit	Standard binomial logit	44.2%	-0.473	0.0%	–
FC-017	sample size	N = 1000	chance-corrected logit	Standard binomial probit	38.3%	-0.243	0.0%	–
FC-018	scale location	Intercept = -1.50	chance-corrected logit	Chance-corrected binomial logit	4.9%	0.000	37.8%	84.4%
FC-018	scale location	Intercept = -1.50	chance-corrected logit	Gaussian identity	16.3%	-0.079	0.0%	–
FC-018	scale location	Intercept = -1.50	chance-corrected logit	Standard binomial logit	20.3%	-0.365	0.0%	–
FC-018	scale location	Intercept = -1.50	chance-corrected logit	Standard binomial probit	19.9%	-0.222	0.0%	–

Table 11: Scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit-problem rate	Apparent rejection rate among flagged fits
FC-019	scale location	Intercept = 1.50	chance-corrected logit	Chance-corrected binomial logit	6.0%	0.000	34.8%	95.5%
FC-019	scale location	Intercept = 1.50	chance-corrected logit	Gaussian identity	11.7%	0.071	0.0%	–
FC-019	scale location	Intercept = 1.50	chance-corrected logit	Standard binomial logit	9.8%	-0.286	36.3%	95.3%
FC-019	scale location	Intercept = 1.50	chance-corrected logit	Standard binomial probit	7.0%	-0.053	0.0%	–
FC-020	trial count	Trials = 5	chance-corrected logit	Chance-corrected binomial logit	4.9%	0.000	21.0%	82.8%
FC-020	trial count	Trials = 5	chance-corrected logit	Gaussian identity	4.5%	-0.036	0.7%	0.0%
FC-020	trial count	Trials = 5	chance-corrected logit	Standard binomial logit	9.0%	-0.473	4.8%	78.3%
FC-020	trial count	Trials = 5	chance-corrected logit	Standard binomial probit	8.7%	-0.243	3.9%	62.7%
FC-021	trial count	Trials = 10	chance-corrected logit	Chance-corrected binomial logit	5.3%	0.000	33.3%	89.8%
FC-021	trial count	Trials = 10	chance-corrected logit	Gaussian identity	5.2%	-0.036	0.0%	–

Table 11: Scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit-problem rate	Apparent rejection rate among flagged fits
FC-021	trial count	Trials = 10	chance-corrected logit	Standard binomial logit	12.2%	-0.473	0.0%	–
FC-021	trial count	Trials = 10	chance-corrected logit	Standard binomial probit	11.8%	-0.243	13.6%	87.0%
FC-022	trial count	Trials = 50	chance-corrected logit	Chance-corrected binomial logit	6.0%	0.000	0.0%	–
FC-022	trial count	Trials = 50	chance-corrected logit	Gaussian identity	6.3%	-0.036	0.0%	–
FC-022	trial count	Trials = 50	chance-corrected logit	Standard binomial logit	19.9%	-0.473	0.0%	–
FC-022	trial count	Trials = 50	chance-corrected logit	Standard binomial probit	17.2%	-0.243	0.0%	–
FC-023	chance level	Chance = 0.25	chance-corrected logit	Chance-corrected binomial logit	5.8%	0.000	40.5%	94.6%
FC-023	chance level	Chance = 0.25	chance-corrected logit	Gaussian identity	6.0%	-0.054	0.0%	–
FC-023	chance level	Chance = 0.25	chance-corrected logit	Standard binomial logit	11.2%	-0.411	0.0%	–
FC-023	chance level	Chance = 0.25	chance-corrected logit	Standard binomial probit	10.6%	-0.228	0.0%	–

Table 11: Scenario-level results for the forced-choice atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction	Fit-problem rate	Apparent rejection rate among flagged fits
FC-024	chance level	Chance = 0.3333	chance-corrected logit	Chance-corrected binomial logit	5.6%	0.000	41.0%	94.5%
FC-024	chance level	Chance = 0.3333	chance-corrected logit	Gaussian identity	6.6%	-0.048	0.0%	0.0%
FC-024	chance level	Chance = 0.3333	chance-corrected logit	Standard binomial logit	13.6%	-0.444	0.0%	–
FC-024	chance level	Chance = 0.3333	chance-corrected logit	Standard binomial probit	12.6%	-0.240	0.0%	–

8.2 Logit versus probit

Table 12: Scenario-level results for the within-family logit-probit atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction
WF-001	paper anchor	Logit-generated manuscript scenario	logit	Binomial logit	5.6%	0.000
WF-001	paper anchor	Logit-generated manuscript scenario	logit	Binomial probit	16.5%	-0.112
WF-002	paper anchor	Probit-generated manuscript scenario	probit	Binomial logit	17.6%	0.216
WF-002	paper anchor	Probit-generated manuscript scenario	probit	Binomial probit	5.0%	0.000
WF-003	main effect surface	logit generated: group x 0.50; condition x 0.50	logit	Binomial logit	4.5%	0.000

Table 12: Scenario-level results for the within-family logit-probit atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction
WF-003	main effect surface	logit generated: group x 0.50; condition x 0.50	logit	Binomial probit	7.4%	-0.048
WF-004	main effect surface	logit generated: group x 1.00; condition x 0.50	logit	Binomial logit	4.9%	0.000
WF-004	main effect surface	logit generated: group x 1.00; condition x 0.50	logit	Binomial probit	10.3%	-0.080
WF-005	main effect surface	logit generated: group x 1.50; condition x 0.50	logit	Binomial logit	4.7%	0.000
WF-005	main effect surface	logit generated: group x 1.50; condition x 0.50	logit	Binomial probit	11.7%	-0.080
WF-006	main effect surface	logit generated: group x 0.50; condition x 1.00	logit	Binomial logit	5.3%	0.000
WF-006	main effect surface	logit generated: group x 0.50; condition x 1.00	logit	Binomial probit	11.8%	-0.080
WF-007	main effect surface	logit generated: group x 1.50; condition x 1.00	logit	Binomial logit	5.6%	0.000
WF-007	main effect surface	logit generated: group x 1.50; condition x 1.00	logit	Binomial probit	11.2%	-0.080
WF-008	main effect surface	logit generated: group x 0.50; condition x 1.50	logit	Binomial logit	5.2%	0.000
WF-008	main effect surface	logit generated: group x 0.50; condition x 1.50	logit	Binomial probit	10.2%	-0.080
WF-009	main effect surface	logit generated: group x 1.00; condition x 1.50	logit	Binomial logit	5.2%	0.000

Table 12: Scenario-level results for the within-family logit-probit atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction
WF-009	main effect surface	logit generated: group x 1.00; condition x 1.50	logit	Binomial probit	10.6%	-0.080
WF-010	main effect surface	logit generated: group x 1.50; condition x 1.50	logit	Binomial logit	5.1%	0.000
WF-010	main effect surface	logit generated: group x 1.50; condition x 1.50	logit	Binomial probit	5.8%	0.000
WF-011	main effect surface	probit generated: group x 0.50; condition x 0.50	probit	Binomial logit	8.5%	0.107
WF-011	main effect surface	probit generated: group x 0.50; condition x 0.50	probit	Binomial probit	5.7%	0.000
WF-012	main effect surface	probit generated: group x 1.00; condition x 0.50	probit	Binomial logit	12.4%	0.162
WF-012	main effect surface	probit generated: group x 1.00; condition x 0.50	probit	Binomial probit	5.2%	0.000
WF-013	main effect surface	probit generated: group x 1.50; condition x 0.50	probit	Binomial logit	13.2%	0.162
WF-013	main effect surface	probit generated: group x 1.50; condition x 0.50	probit	Binomial probit	5.3%	0.000
WF-014	main effect surface	probit generated: group x 0.50; condition x 1.00	probit	Binomial logit	11.2%	0.162
WF-014	main effect surface	probit generated: group x 0.50; condition x 1.00	probit	Binomial probit	4.9%	0.000
WF-015	main effect surface	probit generated: group x 1.50; condition x 1.00	probit	Binomial logit	12.1%	0.162

Table 12: Scenario-level results for the within-family logit-probit atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction
WF-015	main effect surface	probit generated: group x 1.50; condition x 1.00	probit	Binomial probit	5.1%	0.000
WF-016	main effect surface	probit generated: group x 0.50; condition x 1.50	probit	Binomial logit	10.9%	0.162
WF-016	main effect surface	probit generated: group x 0.50; condition x 1.50	probit	Binomial probit	4.8%	0.000
WF-017	main effect surface	probit generated: group x 1.00; condition x 1.50	probit	Binomial logit	10.2%	0.162
WF-017	main effect surface	probit generated: group x 1.00; condition x 1.50	probit	Binomial probit	5.2%	0.000
WF-018	main effect surface	probit generated: group x 1.50; condition x 1.50	probit	Binomial logit	4.4%	0.000
WF-018	main effect surface	probit generated: group x 1.50; condition x 1.50	probit	Binomial probit	4.7%	0.000
WF-019	sample size	logit generated: subjects = 100	logit	Binomial logit	5.5%	0.000
WF-019	sample size	logit generated: subjects = 100	logit	Binomial probit	7.4%	-0.112
WF-020	sample size	logit generated: subjects = 200	logit	Binomial logit	5.6%	0.000
WF-020	sample size	logit generated: subjects = 200	logit	Binomial probit	9.3%	-0.112
WF-021	sample size	logit generated: subjects = 300	logit	Binomial logit	5.7%	0.000
WF-021	sample size	logit generated: subjects = 300	logit	Binomial probit	11.9%	-0.112
WF-022	sample size	logit generated: subjects = 400	logit	Binomial logit	4.4%	0.000

Table 12: Scenario-level results for the within-family logit-probit atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction
WF-022	sample size	logit generated: subjects = 400	logit	Binomial probit	11.8%	-0.112
WF-023	sample size	logit generated: subjects = 1200	logit	Binomial logit	5.3%	0.000
WF-023	sample size	logit generated: subjects = 1200	logit	Binomial probit	25.9%	-0.112
WF-024	sample size	logit generated: subjects = 2000	logit	Binomial logit	4.7%	0.000
WF-024	sample size	logit generated: subjects = 2000	logit	Binomial probit	38.5%	-0.112
WF-025	trial count	logit generated: trials per cell = 5	logit	Binomial logit	4.8%	0.000
WF-025	trial count	logit generated: trials per cell = 5	logit	Binomial probit	8.2%	-0.112
WF-026	trial count	logit generated: trials per cell = 30	logit	Binomial logit	4.4%	0.000
WF-026	trial count	logit generated: trials per cell = 30	logit	Binomial probit	31.0%	-0.112
WF-027	latent location	logit generated: reference intercept = 0	logit	Binomial logit	5.4%	0.000
WF-027	latent location	logit generated: reference intercept = 0	logit	Binomial probit	38.2%	0.175
WF-028	latent location	logit generated: reference intercept = 0.75	logit	Binomial logit	5.1%	0.000
WF-028	latent location	logit generated: reference intercept = 0.75	logit	Binomial probit	8.2%	0.059
WF-029	latent location	logit generated: reference intercept = 2.25	logit	Binomial logit	5.6%	0.000
WF-029	latent location	logit generated: reference intercept = 2.25	logit	Binomial probit	44.1%	-0.186

Table 12: Scenario-level results for the within-family logit-probit atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction
WF-030	icc	logit generated: target ICC = 0	logit	Binomial logit	4.9%	0.000
WF-030	icc	logit generated: target ICC = 0	logit	Binomial probit	78.7%	-0.112
WF-031	icc	logit generated: target ICC = 0.15	logit	Binomial logit	5.5%	0.000
WF-031	icc	logit generated: target ICC = 0.15	logit	Binomial probit	37.0%	-0.112
WF-032	icc	logit generated: target ICC = 0.45	logit	Binomial logit	5.9%	0.000
WF-032	icc	logit generated: target ICC = 0.45	logit	Binomial probit	9.2%	-0.112
WF-033	sample size	probit generated: subjects = 100	probit	Binomial logit	6.6%	0.216
WF-033	sample size	probit generated: subjects = 100	probit	Binomial probit	5.3%	0.000
WF-034	sample size	probit generated: subjects = 200	probit	Binomial logit	8.8%	0.216
WF-034	sample size	probit generated: subjects = 200	probit	Binomial probit	5.5%	0.000
WF-035	sample size	probit generated: subjects = 300	probit	Binomial logit	11.5%	0.216
WF-035	sample size	probit generated: subjects = 300	probit	Binomial probit	5.6%	0.000
WF-036	sample size	probit generated: subjects = 400	probit	Binomial logit	11.6%	0.216
WF-036	sample size	probit generated: subjects = 400	probit	Binomial probit	4.5%	0.000
WF-037	sample size	probit generated: subjects = 1200	probit	Binomial logit	28.1%	0.216
WF-037	sample size	probit generated: subjects = 1200	probit	Binomial probit	5.5%	0.000
WF-038	sample size	probit generated: subjects = 2000	probit	Binomial logit	41.5%	0.216

Table 12: Scenario-level results for the within-family logit-probit atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction
WF-038	sample size	probit generated: subjects = 2000	probit	Binomial probit	5.6%	0.000
WF-039	trial count	probit generated: trials per cell = 5	probit	Binomial logit	7.9%	0.216
WF-039	trial count	probit generated: trials per cell = 5	probit	Binomial probit	4.8%	0.000
WF-040	trial count	probit generated: trials per cell = 30	probit	Binomial logit	32.2%	0.216
WF-040	trial count	probit generated: trials per cell = 30	probit	Binomial probit	4.6%	0.000
WF-041	latent location	probit generated: reference intercept = 0	probit	Binomial logit	39.9%	-0.424
WF-041	latent location	probit generated: reference intercept = 0	probit	Binomial probit	5.7%	0.000
WF-042	latent location	probit generated: reference intercept = 0.75	probit	Binomial logit	8.0%	-0.108
WF-042	latent location	probit generated: reference intercept = 0.75	probit	Binomial probit	4.7%	0.000
WF-043	latent location	probit generated: reference intercept = 2.25	probit	Binomial logit	46.2%	0.520
WF-043	latent location	probit generated: reference intercept = 2.25	probit	Binomial probit	4.7%	0.000
WF-044	icc	probit generated: target ICC = 0	probit	Binomial logit	82.3%	0.216
WF-044	icc	probit generated: target ICC = 0	probit	Binomial probit	5.0%	0.000
WF-045	icc	probit generated: target ICC = 0.15	probit	Binomial logit	40.9%	0.216
WF-045	icc	probit generated: target ICC = 0.15	probit	Binomial probit	4.8%	0.000

Table 12: Scenario-level results for the within-family logit-probit atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction
WF-046	icc	probit generated: target ICC = 0.45	probit	Binomial logit	8.6%	0.216
WF-046	icc	probit generated: target ICC = 0.45	probit	Binomial probit	5.4%	0.000

8.3 Sum scores

Table 13: Scenario-level results for the sum-score atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction
SS-001	paper anchor	Lower range	graded logistic latent scale	Gaussian-probit bounded score	7.9%	-0.056
SS-001	paper anchor	Lower range	graded logistic latent scale	Latent generating scale	4.9%	0.000
SS-001	paper anchor	Lower range	graded logistic latent scale	Observed sum-score identity	72.8%	-2.034
SS-002	paper anchor	Middle range	graded logistic latent scale	Gaussian-probit bounded score	3.5%	-0.010
SS-002	paper anchor	Middle range	graded logistic latent scale	Latent generating scale	5.6%	0.000
SS-002	paper anchor	Middle range	graded logistic latent scale	Observed sum-score identity	8.6%	-0.696

Table 13: Scenario-level results for the sum-score atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction
SS-003	paper anchor	Upper range	graded logistic latent scale	Gaussian-probit bounded score	4.8%	0.027
SS-003	paper anchor	Upper range	graded logistic latent scale	Latent generating scale	4.5%	0.000
SS-003	paper anchor	Upper range	graded logistic latent scale	Observed sum-score identity	37.4%	1.570
SS-004	main effect surface	x effect x 0.50; group effect x 0.50	graded logistic latent scale	Gaussian-probit bounded score	4.7%	-0.001
SS-004	main effect surface	x effect x 0.50; group effect x 0.50	graded logistic latent scale	Latent generating scale	4.9%	0.000
SS-004	main effect surface	x effect x 0.50; group effect x 0.50	graded logistic latent scale	Observed sum-score identity	4.8%	-0.096
SS-005	main effect surface	x effect x 1.00; group effect x 0.50	graded logistic latent scale	Gaussian-probit bounded score	3.3%	-0.002
SS-005	main effect surface	x effect x 1.00; group effect x 0.50	graded logistic latent scale	Latent generating scale	4.3%	0.000
SS-005	main effect surface	x effect x 1.00; group effect x 0.50	graded logistic latent scale	Observed sum-score identity	2.6%	-0.180
SS-006	main effect surface	x effect x 1.50; group effect x 0.50	graded logistic latent scale	Gaussian-probit bounded score	3.2%	-0.004

Table 13: Scenario-level results for the sum-score atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction
SS-006	main effect surface	x effect x 1.50; group effect x 0.50	graded logistic latent scale	Latent generating scale	4.8%	0.000
SS-006	main effect surface	x effect x 1.50; group effect x 0.50	graded logistic latent scale	Observed sum-score identity	2.4%	-0.242
SS-007	main effect surface	x effect x 0.50; group effect x 1.00	graded logistic latent scale	Gaussian-probit bounded score	4.6%	-0.005
SS-007	main effect surface	x effect x 0.50; group effect x 1.00	graded logistic latent scale	Latent generating scale	4.5%	0.000
SS-007	main effect surface	x effect x 0.50; group effect x 1.00	graded logistic latent scale	Observed sum-score identity	6.4%	-0.370
SS-008	main effect surface	x effect x 1.50; group effect x 1.00	graded logistic latent scale	Gaussian-probit bounded score	3.5%	-0.015
SS-008	main effect surface	x effect x 1.50; group effect x 1.00	graded logistic latent scale	Latent generating scale	5.4%	0.000
SS-008	main effect surface	x effect x 1.50; group effect x 1.00	graded logistic latent scale	Observed sum-score identity	10.5%	-0.940
SS-009	main effect surface	x effect x 0.50; group effect x 1.50	graded logistic latent scale	Gaussian-probit bounded score	4.9%	-0.011
SS-009	main effect surface	x effect x 0.50; group effect x 1.50	graded logistic latent scale	Latent generating scale	5.6%	0.000
SS-009	main effect surface	x effect x 0.50; group effect x 1.50	graded logistic latent scale	Observed sum-score identity	13.3%	-0.786

Table 13: Scenario-level results for the sum-score atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction
SS-010	main effect surface	x effect x 1.00; group effect x 1.50	graded logistic latent scale	Gaussian-probit bounded score	4.5%	-0.022
SS-010	main effect surface	x effect x 1.00; group effect x 1.50	graded logistic latent scale	Latent generating scale	5.1%	0.000
SS-010	main effect surface	x effect x 1.00; group effect x 1.50	graded logistic latent scale	Observed sum-score identity	30.0%	-1.481
SS-011	main effect surface	x effect x 1.50; group effect x 1.50	graded logistic latent scale	Gaussian-probit bounded score	3.5%	-0.036
SS-011	main effect surface	x effect x 1.50; group effect x 1.50	graded logistic latent scale	Latent generating scale	4.9%	0.000
SS-011	main effect surface	x effect x 1.50; group effect x 1.50	graded logistic latent scale	Observed sum-score identity	42.9%	-2.008
SS-012	sample size	N = 75	graded logistic latent scale	Gaussian-probit bounded score	3.1%	-0.010
SS-012	sample size	N = 75	graded logistic latent scale	Latent generating scale	5.3%	0.000
SS-012	sample size	N = 75	graded logistic latent scale	Observed sum-score identity	3.6%	-0.696
SS-013	sample size	N = 150	graded logistic latent scale	Gaussian-probit bounded score	4.0%	-0.010

Table 13: Scenario-level results for the sum-score atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction
SS-013	sample size	N = 150	graded logistic latent scale	Latent generating scale	5.1%	0.000
SS-013	sample size	N = 150	graded logistic latent scale	Observed sum-score identity	4.1%	-0.696
SS-014	sample size	N = 225	graded logistic latent scale	Gaussian-probit bounded score	3.4%	-0.010
SS-014	sample size	N = 225	graded logistic latent scale	Latent generating scale	5.4%	0.000
SS-014	sample size	N = 225	graded logistic latent scale	Observed sum-score identity	4.5%	-0.696
SS-015	sample size	N = 300	graded logistic latent scale	Gaussian-probit bounded score	4.3%	-0.010
SS-015	sample size	N = 300	graded logistic latent scale	Latent generating scale	4.8%	0.000
SS-015	sample size	N = 300	graded logistic latent scale	Observed sum-score identity	5.4%	-0.696
SS-016	sample size	N = 450	graded logistic latent scale	Gaussian-probit bounded score	3.5%	-0.010
SS-016	sample size	N = 450	graded logistic latent scale	Latent generating scale	5.2%	0.000
SS-016	sample size	N = 450	graded logistic latent scale	Observed sum-score identity	7.1%	-0.696

Table 13: Scenario-level results for the sum-score atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction
SS-017	sample size	N = 1200	graded logistic latent scale	Gaussian-probit bounded score	3.4%	-0.010
SS-017	sample size	N = 1200	graded logistic latent scale	Latent generating scale	4.5%	0.000
SS-017	sample size	N = 1200	graded logistic latent scale	Observed sum-score identity	12.1%	-0.696
SS-018	scale location	Threshold shift = -2.00	graded logistic latent scale	Gaussian-probit bounded score	6.2%	0.038
SS-018	scale location	Threshold shift = -2.00	graded logistic latent scale	Latent generating scale	6.0%	0.000
SS-018	scale location	Threshold shift = -2.00	graded logistic latent scale	Observed sum-score identity	56.8%	1.877
SS-019	scale location	Threshold shift = 2.00	graded logistic latent scale	Gaussian-probit bounded score	11.9%	-0.072
SS-019	scale location	Threshold shift = 2.00	graded logistic latent scale	Latent generating scale	4.6%	0.000
SS-019	scale location	Threshold shift = 2.00	graded logistic latent scale	Observed sum-score identity	82.1%	-1.993
SS-020	item count	Items = 5	graded logistic latent scale	Gaussian-probit bounded score	3.5%	-0.013

Table 13: Scenario-level results for the sum-score atlas.

ID	Slice	Scenario	Generating link/scale	Fitted model/scale	Rejection rate	Deterministic pseudo-interaction
SS-020	item count	Items = 5	graded logistic latent scale	Latent generating scale	4.7%	0.000
SS-020	item count	Items = 5	graded logistic latent scale	Observed sum-score identity	7.4%	-0.384
SS-021	item count	Items = 15	graded logistic latent scale	Gaussian-probit bounded score	3.7%	-0.008
SS-021	item count	Items = 15	graded logistic latent scale	Latent generating scale	5.2%	0.000
SS-021	item count	Items = 15	graded logistic latent scale	Observed sum-score identity	9.1%	-1.162
SS-022	latent dispersion	Latent SD = 0.70	graded logistic latent scale	Gaussian-probit bounded score	3.9%	-0.013
SS-022	latent dispersion	Latent SD = 0.70	graded logistic latent scale	Latent generating scale	4.2%	0.000
SS-022	latent dispersion	Latent SD = 0.70	graded logistic latent scale	Observed sum-score identity	13.4%	-0.831
SS-023	latent dispersion	Latent SD = 1.30	graded logistic latent scale	Gaussian-probit bounded score	3.3%	-0.007
SS-023	latent dispersion	Latent SD = 1.30	graded logistic latent scale	Latent generating scale	4.7%	0.000

SS-023	latent dispersion	Latent SD = 1.30	graded logistic latent scale	Observed sum-score identity	5.5%	-0.563